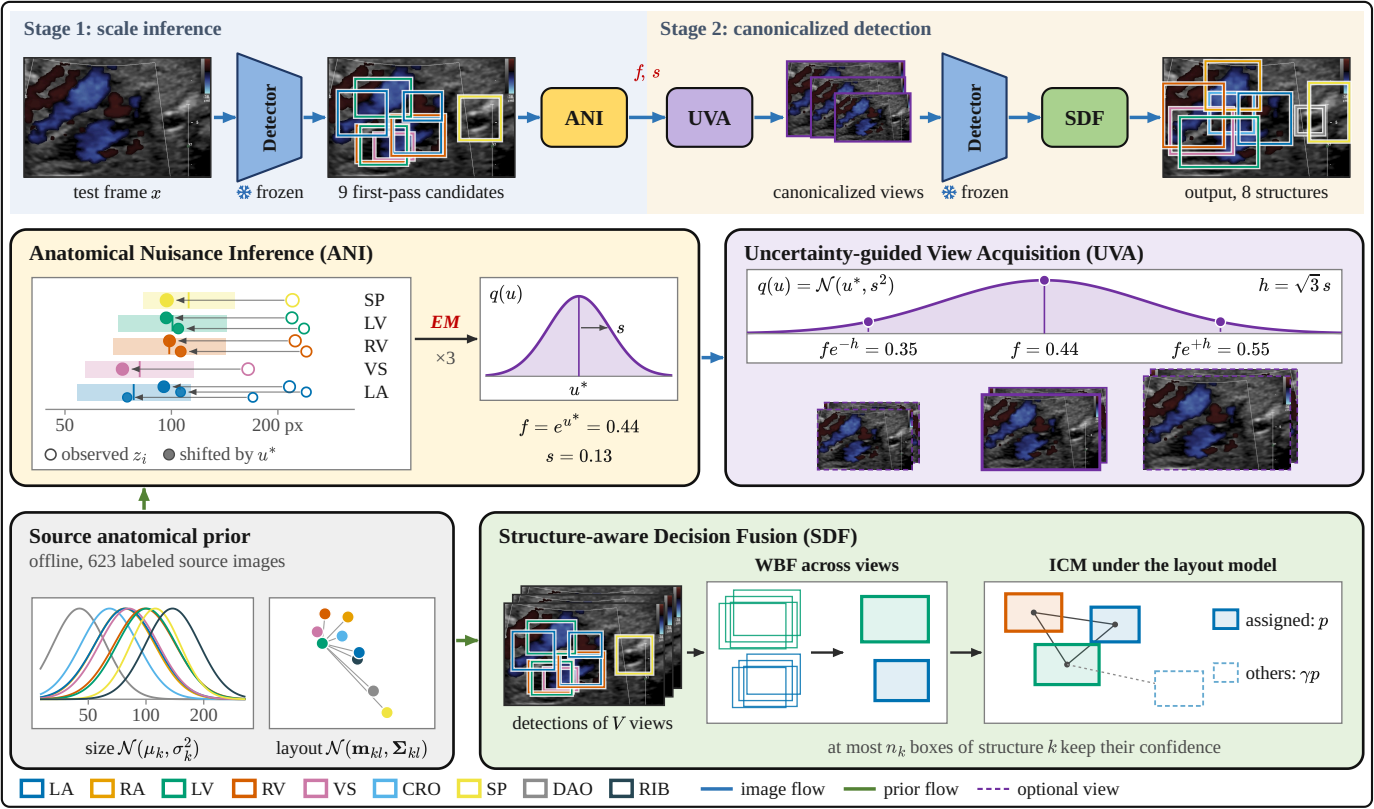

Stage 1: scale inference
Stage 2: canonicalized detection
test frame $x$
Detector
❄ frozen
9 first-pass candidates
ANI
$f$, $s$
UVA
canonicalized views
Detector
❄ frozen
SDF
output, 8 structures
Anatomical Nuisance Inference (ANI)
SP
LV
RV
VS
LA
50
100
200 px
observed $z_i$
shifted by $u^*$
EM
×3
$q(u)$
$s$
$u^*$
$f = e^{u^*} = 0.44$
$s = 0.13$
Uncertainty-guided View Acquisition (UVA)
$q(u) = \mathcal{N}(u^*, s^2)$
$h = \sqrt{3}\,s$
$fe^{-h} = 0.35$
$f = 0.44$
$fe^{+h} = 0.55$
Source anatomical prior
offline, 623 labeled source images
50
100
200
size $\mathcal{N}(\mu_k, \sigma_k^2)$
layout $\mathcal{N}(\mathbf{m}_{kl}, \mathbf{\Sigma}_{kl})$
Structure-aware Decision Fusion (SDF)
WBF across views
ICM under the layout model
assigned: $p$
others: $\gamma p$
detections of $V$ views
at most $n_k$ boxes of structure $k$ keep their confidence
LA
RA
LV
RV
VS
CRO
SP
DAO
RIB
image flow
prior flow
optional view